

Ultra-broadband near- to mid-infrared electro-optic modulator on thin-film lithium niobate

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The escalating capacity limitations of conventional near-infrared telecommunication bands have spurred urgent investigations into wide-band optical communication systems spanning from the near-infrared to mid-infrared regimes. This has motivated the development of optical components combining broadband bandwidth with high-speed operation. The state-of-the-art modulators face challenges in achieving broad operational bandwidth due to waveguide dispersion and velocity mismatch. Here we demonstrate a thin-film lithium niobate (TFLN) electro-optic (EO) modulator with an unprecedented 800-nm operational bandwidth, covering the full O-U telecom bands and extending into the 2- μ m regime. The TFLN modulator exhibits >67 GHz EO bandwidth across O-U bands (~100 GHz at O-/S-/C-/L-bands) and >50 GHz at 2- μ m band. It enables single-lane exceeding 240 Gbps PAM-4 transmission across O-U bands and a record 170 Gbps PAM-4 transmission at 2- μ m band. This breakthrough establishes TFLN as a compelling platform for multispectral photonics, bridging conventional telecom infrastructure with emerging 2- μ m technologies for next-generation optical communications.

The relentless growth of big data, cloud computing, and artificial intelligence has escalated global data traffic to unprecedented levels, challenging the capacity limits of conventional single-mode fiber (SMF)-based optical communication systems¹. To address this, two complementary strategies have emerged: maximizing the utilization of existing spectral resources and exploring new optical communication windows beyond traditional O- and C-bands^{2,3}. Recent advances in hollow-core photonic bandgap fibers (HC-PBGFs) have unveiled ultra-broad low-loss transmission windows spanning 1240–1940 nm^{4–6}, covering the O-U bands and the newly proposed 2- μ m band. At the

meantime, innovations in amplifier technologies—ranging from rare-earth-doped fibers to the newly demonstrated integrated optical parametric amplifiers—deliver broadband gain across these extended spectral regions^{7–9}. Together, these developments enable a transformative ultra-wide communication spectrum that seamlessly integrates conventional telecom bands with the emerging 2- μ m waveband. Beyond revolutionizing optical networks, this expanded spectral range also holds promise for applications such as quantum photonics, precision metrology, and biomedical imaging^{10–12}. However, the realization of end-to-end optical systems spanning these wavelengths

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remains hindered by a critical gap: the absence of high-speed broadband optical transmitters and photodetectors (PDs).

Integrated PDs for the 2- μm band have achieved bandwidths exceeding 40 GHz, inherently supporting high-speed optical detection across the O-U bands^{13–18}. In contrast, the optical bandwidth of their modulator counterparts remains constrained. Existing integrated modulation platforms, such as silicon and germanium modulators relying on free-carrier effects^{19–23}, face inherent challenges at longer wavelengths due to waveguide dispersion and carrier velocity mismatch. These limitations confine existing modulators to single-band operations, hindering their adaptation to emerging multi-band or ultrabroadband communication systems. Thin-film lithium niobate (TFLN) has emerged as a uniquely promising platform for optical modulation, combining advantages such as ultra-broad optical transparency (visible to mid-infrared spectrum), large electro-optic (EO) coefficients enabled by the Pockels effect, low optical loss, and compatibility with CMOS voltages enables modulators with unmatched bandwidth and efficiency^{24–29}. Recent milestones include TFLN-based coherent transmission at 1.96 Tb/s³⁰ and compact packaged modules³¹, underscoring its potential for next-generation networks. Although previous TFLN modulators have operated across discrete wavelengths from visible to 2- μm bands^{24,32–35}, achieving seamless ultrabroadband operation for both the O-U telecom bands and the 2- μm bands—while maintaining the high EO bandwidth—remains an unresolved challenge, which is due to the intrinsic optical and EO bandwidth limitations of conventional modulator designs.

Here, we co-design the passive components and modulation electrodes and demonstrate a high-speed TFLN EO modulator with a record-breaking continuous operational range of 1260–2060 nm, seamlessly bridging the O-U bands and the 2- μm band. This broadband capability is enabled by leveraging adiabatic mode evolution engineering in critical passive optical components—such as optimized optical splitters and spot-size converters (SSCs). By integrating the broadband optical splitters and SSCs with high-performance traveling-wave electrodes, our modulator achieves a 3-dB EO bandwidth exceeding 67 GHz across the O-U bands and 50 GHz in the 2- μm band—the highest reported for this spectral region. Furthermore, the modulator achieves a 3-dB EO bandwidth of ~100 GHz in the O-, S-, C-, and L-bands (1310 nm, 1485 nm, 1550 nm, 1590 nm). The modulator exhibits exceptional efficiency, with V_{π}/L values of 1.92, 2.48, 2.61, 2.74, and 3.94 V·cm at 1310, 1485, 1550, 1590, and 2000 nm, respectively. For the first time with a single modulator, we demonstrate transmission in the O-, E-, S-, C-, L-, U-, and 2- μm bands; data rates reached 170/170/170/180/180/170/150 Gbps (On-Off Keying, OOK) and 260/260/260/280/280/240/170 Gbps (4-level Pulse Amplitude Modulation, PAM-4), respectively, with bit error rate (BER) all below the hard-decision forward error correction (HD-FEC) threshold (3.8×10^{-3}). This breakthrough paves the way for universal optical transmitters capable of unifying fragmented spectral resources into a single ultra-broadband communication infrastructure, with transformative potential for next-generation optical networks, quantum systems, and sensing technologies.

Results

Design of the broadband TFLN modulator

As shown in Fig. 1a, the proposed extended full-spectrum optical communication system leverages TFLN technology to realize broadband optical transmitters spanning the O-band to the 2- μm wavelength range. This approach integrates the TFLN platform (NanoLN) with existing multi-wavelength laser sources and multi-band optical amplifiers. At the transmitter, a single full-spectrum TFLN EO modulator converts electrical signals—including OOK and advanced modulation formats such as PAM-4—into optical signals across the entire spectral range from the O-band to the 2- μm band. The EOMs are fabricated on a 300-nm-thick x-cut TFLN layer bonded to a 4.7- μm

thermally grown silicon dioxide substrate with a silicon base. As illustrated in the inset of Fig. 1a, the waveguide architecture features a 120-nm slab height and a sidewall angle of $\theta = 60^\circ$. Figure 1b displays the microscopy images of the fabricated EO modulator, which comprises two 3-dB power splitters, a 9-mm-long EO Mach-Zehnder interferometer modulation section, and a thermo-optic phase shifter, and two SSCs as the edge coupler for light coupling. The SEM images of an SSC and a power splitter are shown in the inset of Fig. 1b.

To achieve ultra-broadband operating bandwidth for practical optical packaging, the edge coupler was designed based on broadband SSC, which is illustrated in Fig. 2a, b. The structure comprises LN bi-layer tapers and a silicon oxynitride (SiON) ridge waveguide. At the chip edge, the SiON waveguide interfaces with an ultra-high numerical-aperture (UHNA4) fiber. The fundamental transverse electric (TE) mode transitions adiabatically from the SiON waveguide to the LN waveguide via tapered thickness modulation. The refractive index of SiON ($n = 1.54$) lies between that of LN ($n \approx 2.2$) and the silicon dioxide buried oxide (SiO_2 BOX, $n \approx 1.45$), enabling optical confinement within the waveguide while suppressing leakage into the BOX. Achieving low coupling loss (<1 dB) across the 1260–2060 nm spectral range requires precise matching between the mode field diameter (MFD) of the SiON waveguide and the coupled fiber. The MFD of the UHNA4 fiber increases with wavelength, for example, it is 3.3 μm at 1310 nm, 4.0 μm at 1550 nm, and 5.3 μm at 2000 nm. For a SiON rib waveguide with a rib height of 3.2 μm and slab height of 1 μm , the wavelength-dependent coupling loss with varying the width of the SiON rib waveguide are presented in Fig. 2c. Narrower rib widths minimize coupling loss at shorter wavelengths (e.g., 1310 to 1480 nm), while broader widths improve performance at longer wavelengths (e.g., 1680–2000 nm). However, an anomalous variation in coupling loss occurs at 1310, 1410, and 1480 nm when the rib width exceeds 4.8 μm , caused by resonant fluctuations from higher-order mode excitation. Specifically, increased rib width enhances susceptibility to higher-order modes at shorter wavelengths, inducing resonances that disrupt coupling efficiency. Notably, at 1550 nm, the SSC exhibits consistent low coupling loss across different rib widths. Below 1550 nm, coupling loss increases with rib width, while the opposite trend is observed above 1550 nm. To balance high coupling efficiency across O- to U-bands (with prioritized performance in the C-band) and maintain acceptable loss at longer wavelengths, the SiON rib width was optimized to 4.4 μm . This design also demonstrates robust tolerance to dimensional variations in the SiON waveguide cross-section.

The SSC parameters were optimized through a dual focus: maximizing the taper length to ensure adiabatic mode evolution and minimizing the tip width to enhance coupling efficiency, while maintaining fabrication feasibility. The design details of the SSC including the influence of tip widths to the coupling loss are discussed in Supplementary Note 1. The finalized SSC parameters are summarized in Table 1. Figure 2d presents the calculated transmission spectrum across 1200–2100 nm, with transmission efficiencies of 91.4%, 92.2%, and 88.4% at 1310 nm, 1550 nm, and 2000 nm, respectively. Notably, the simulation framework omitted intrinsic optical absorption losses in plasma-deposited silicon oxynitride (SiO_xN_y). Minor spectral ripples below 1500 nm originate from the reflection-induced interference in the sudden height change of LN waveguide, and the intensity fluctuation of the higher-order modes in the SiON waveguide, as revealed by mode field simulations.

The full spectrum 3-dB power splitter is based on TFLN adiabatic waveguide, with light is adiabatically coupled from the input waveguide to two symmetric output waveguides through three tapers. This design leverages adiabatic coupling principles, enabling uniform 3-dB power splitting over an ultrawide wavelength range provided that adiabatic conditions are satisfied. We employ this broadband power-splitting scheme into the TFLN modulator to construct the light interference structure. Further design details of the 3-dB power splitter

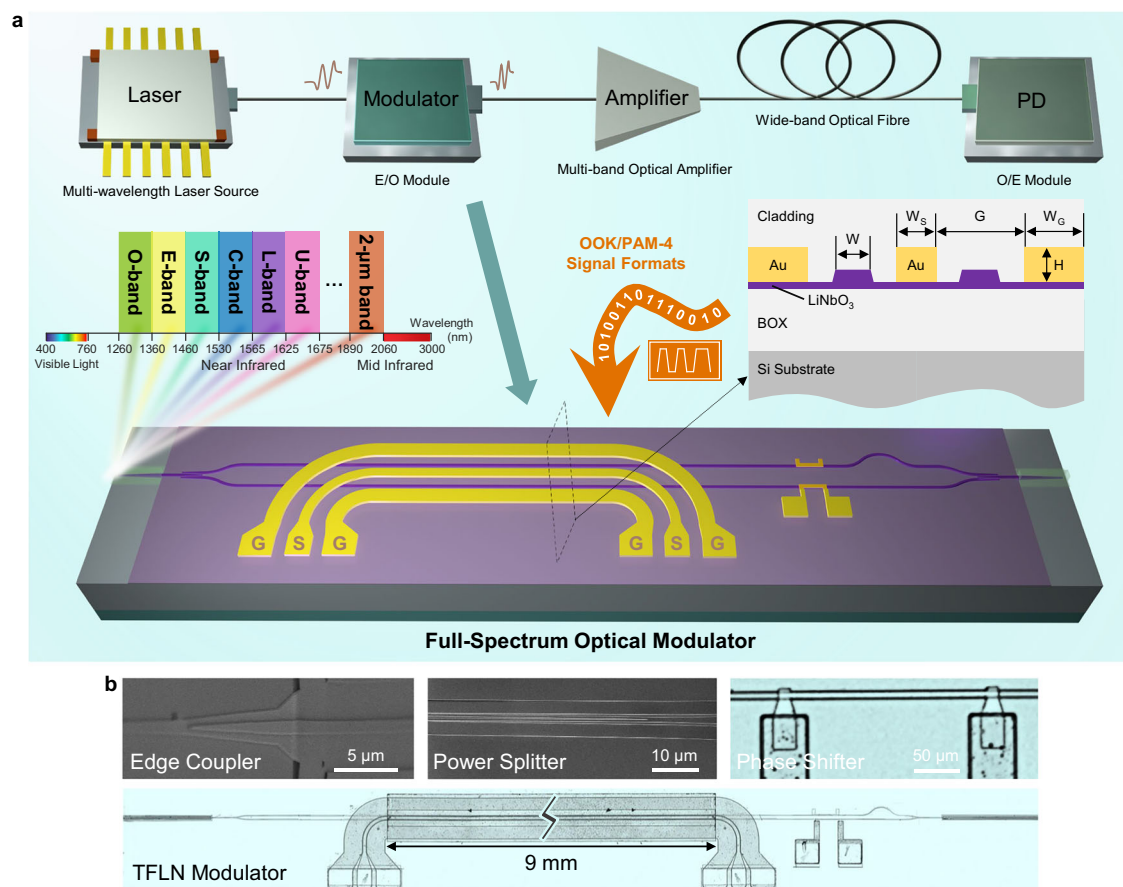


Fig. 1 | Full-spectrum optical modulator. **a** Concept of extended full spectrum optical communication through broadband thin-film lithium niobate (TFLN) optical modulator, multi-wavelength laser sources, multi-band optical amplifiers, wide-

band optical fiber and PDs. Inset: cross-sectional view of the modulator region.

b Microscopy images of the fabricated LN Mach-Zehnder electro-optic modulator (EOM). Inset: SEM images of the spot-size converter (SSC) and power splitter.

are provided in ref.³⁶. For shorter wavelengths (e.g., O-band), waveguides thicker than 300 nm impose stricter adiabatic mode evolution requirements, which would degrade the 3-dB power splitter performance. While the reduced electro-optic interaction at this thickness increases the half-wave voltage, this trade-off was essential to achieve a functional bandwidth spanning 1260–2050 nm. As shown in Fig. 2e, the calculated transmission for each splitter is 48.2%, 49.6%, and 44.6% at 1310, 1550, and 2000 nm, respectively, confirming robust performance across the targeted spectrum.

Outside the modulation region, the waveguides are engineered to support single-mode operation at 1310 nm, ensuring broadband compatibility across the O-/C- and 2-μm bands. This design suppresses higher-order mode excitation over the 1260–2060 nm range, thereby preventing EO performance degradation in the modulation region. Finite-difference eigenmode (FDE) simulations indicate that single-mode operation requires a waveguide top width of ~0.8 μm and therefore the waveguide width was set as $W = 0.8 \mu\text{m}$ in the regions of SSC, power splitter, and bend waveguide. However, this narrow width reduces optical mode confinement, increasing propagation losses due to stronger mode overlap with waveguide sidewalls and cladding, particularly at longer wavelengths. Additionally, the constrained geometry elevates the half-wave voltage requirement. To balance these trade-offs, we implement an adiabatic taper proximal to the 3-dB power splitter, expanding the waveguide width to $W = 3.5 \mu\text{m}$ within the modulation region. The details of the waveguide width optimization are discussed in Supplementary Note 2. A FDE solver was employed to numerically analyze the coupled electrical-optical modal properties of the device architecture. The electrode gap (G) is defined

as the horizontal separation between the ground plane and the signal electrode. As shown in Fig. 2f, we calculated the half-wave voltage-length product ($V_{\pi}L$) and quantified the metal-induced absorption loss, both parameterized as functions of G at wavelengths of 1310, 1550, and 2000 nm. The results reveal an inverse relationship between device modulation efficiency ($V_{\pi}L$) and optical loss performance, necessitating tailored selection of G to prioritize either metric for specific applications. For this work, an electrode gap of $G = 6 \mu\text{m}$ was chosen to achieve an optimal trade-off point, simultaneously ensuring strong modulation performance while limiting absorption loss to moderate levels.

The traveling-wave electrodes feature a push-pull configuration based on a coplanar waveguide (CPW) structure. The signal and ground electrode widths are designed as $W_S = 25 \mu\text{m}$ and $W_G = 150 \mu\text{m}$, respectively, with gold (Au) electrodes of height $H = 1 \mu\text{m}$ and a 2-μm-thick SiO_2 cladding layer. The characteristic impedance (Z_0) of the CPW is approximately 42Ω , as illustrated in the inset of Fig. 2g. A load resistor of $Z_L = 38 \Omega$ is used, which is intentionally lower than Z_0 to mitigate the low-frequency EO S21 roll-off, thereby slightly enhancing the operational bandwidth³⁷. Design details regarding the traveling-wave electrode impedance can be found in Supplementary Note 3. The optical group indices are 2.18, 2.13, and 2.04 at wavelengths (λ) of 1310 nm, 1550 nm, and 2000 nm, respectively (dashed lines in Fig. 2g). The optical group index at the C-band is aligned with the radio frequency (RF) effective index, resulting in a trade-off in EO bandwidth performance for the 2-μm band. The RF effective index is designed as 2.13 at 50 GHz. For $\lambda = 2000 \text{ nm}$, the residual index mismatch leads to a theoretical 3-dB bandwidth of approximately 80 GHz for a 9-mm-long,

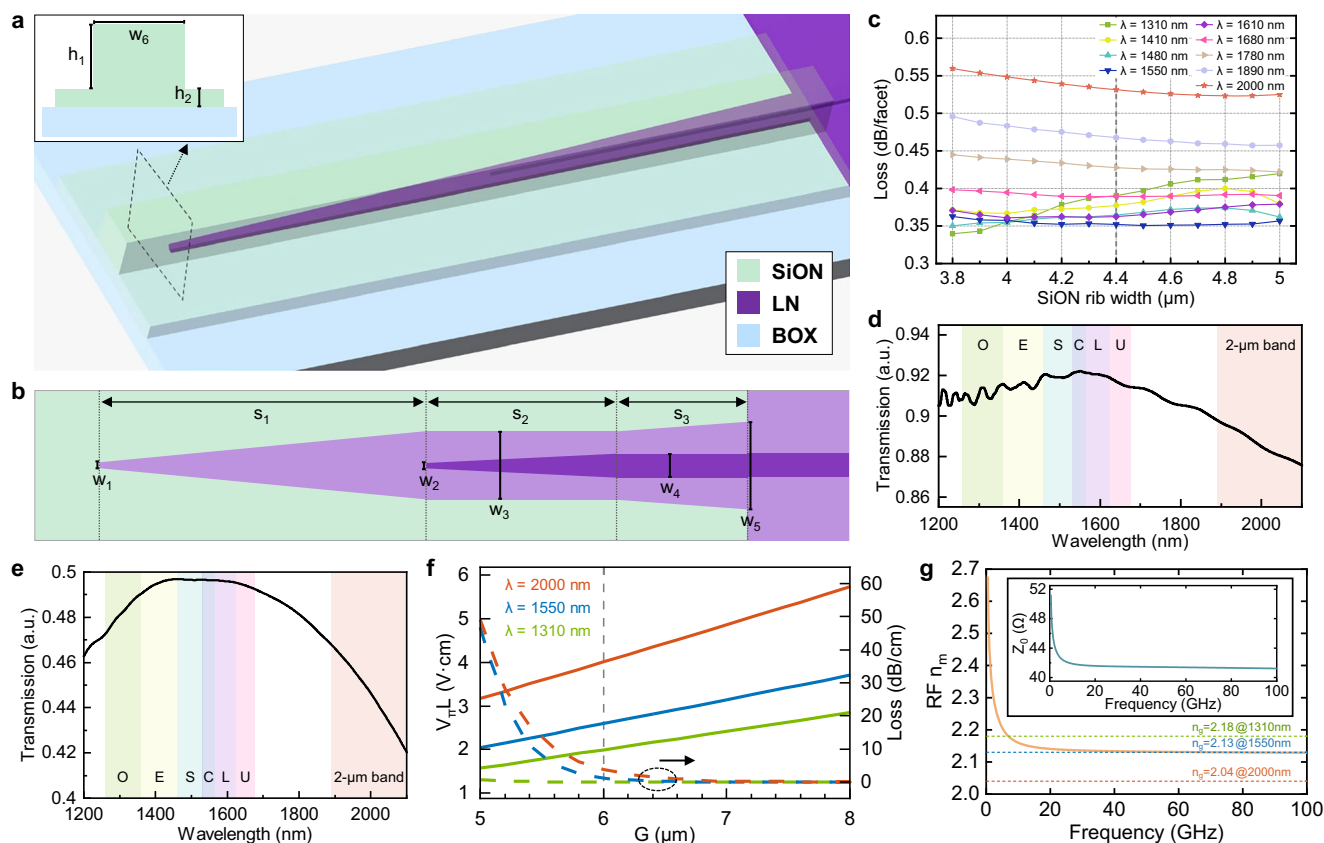


Fig. 2 | Design of the thin-film lithium niobate (TFLN) modulator. a Schematic structure of the spot-size converter (SSC). The inset shows the dimension parameters of the SiON waveguide. **b** Dimension parameters of the LN bi-layer tapers. **c** Dependence of the coupling loss on SiON rib width. **d** Simulated transmission spectrum of the SSC. **e** Simulated transmission spectrum of the 3-dB power splitter.

f Simulated $V_\pi L$ and metal absorption loss as a function of G . **g** Simulated frequency-dependent results of radio frequency (RF) effective index n_m . The three dashed lines correspond to the optical group index n_g . The inset shows the simulated characteristic impedance Z_0 .

Table 1 | Parameters of the SSC

Parameter	w_1	w_2	w_3	w_4	w_5	w_6	s_1	s_2	s_3	h_1	h_2
Value (μm)	0.2	0.2	2.1	0.8	5	4.4	250	150	100	3.2	1

w_1 to w_6 are the widths, s_1 to s_3 are the lengths of the tapered waveguides, h_1 to h_2 are the heights of SiON layers, as shown in Fig. 2a, b.

impedance-matched, lossless modulator. Design details of the broad bandwidth traveling-wave electrodes can be found in Supplementary Note 3. The thermo-optic (TO) phase shifter is employed to establish the optimal operating bias point. By adopting TO tuning instead of EO tuning via bias tees on the traveling-wave electrodes, this approach not only simplifies packaging and testing procedures but also significantly improves the stability of the bias point by avoiding the LN DC bias drift problem caused by the DC bias voltage under EO bias conditions. Detailed design parameters and experimental results of the TO phase shifter used in our modulator can be found in supplementary Note 4.

Measurement and analysis

The fabrication details of TFLN modulator is given in “Methods”. The coupling losses of the SSC were characterized using a reference waveguide coupled to a UHNA4 fiber. Refractive index-matching oil (index = 1.46) was applied at the fiber-chip interface to minimize reflections. Coupling losses were measured using tunable lasers for the 1260–1680 nm bands, and a custom-built 2- μm amplified spontaneous emission (ASE) source paired with an optical spectrum analyzer (Yokogawa AQ6375E) for the 1920–2060 nm range. Using a polarization controller (PC), the measured TE mode coupling losses are

0.78 dB/facet at 1310 nm, 1.18 dB/facet at 1450 nm, 1.73 dB/facet at 1485 nm, 0.69 dB/facet at 1550 nm, 0.56 dB/facet at 1590 nm, 0.69 dB/facet at 1653 nm, 0.82 dB/facet at 1970 nm, and 2.21 dB/facet at 2000 nm, as shown in Fig. 3a. The slightly higher losses at shorter and longer wavelengths are attributed to minor mode mismatch between the UHNA4 fiber and the SiON waveguide. Notably, three pronounced loss peaks near 1510 nm and 2000 nm correlate with intrinsic N-H infrared absorption bands in the plasma-deposited SiON layer, consistent with prior reports of N-H bond absorption^{38–41}. The slight loss rise of 1370–1410 nm can be attributed to the O-H bond during the plasma deposition of SiON layer⁴². The normalized transmission spectra of the TFLN modulator for the 1260–1680 nm and 1920–2060 nm spectral region can be found in Supplementary Note 5. Following characterization of the modulator’s total optical losses, the on-chip insertion losses (ILs) were deduced as 1.2 dB, 2.8 dB, and 5.8 dB at wavelengths of 1310 nm, 1550 nm, and 1970 nm, respectively. A test structure comprising 16 cascaded 3-dB power splitters was used to measure the ILs of individual splitters, yielding values of 0.23 dB, 0.11 dB, and 0.42 dB at 1310 nm, 1550 nm, and 1970 nm, closely matching simulations. After accounting for contributions from the power splitters and waveguide propagation losses, the primary source

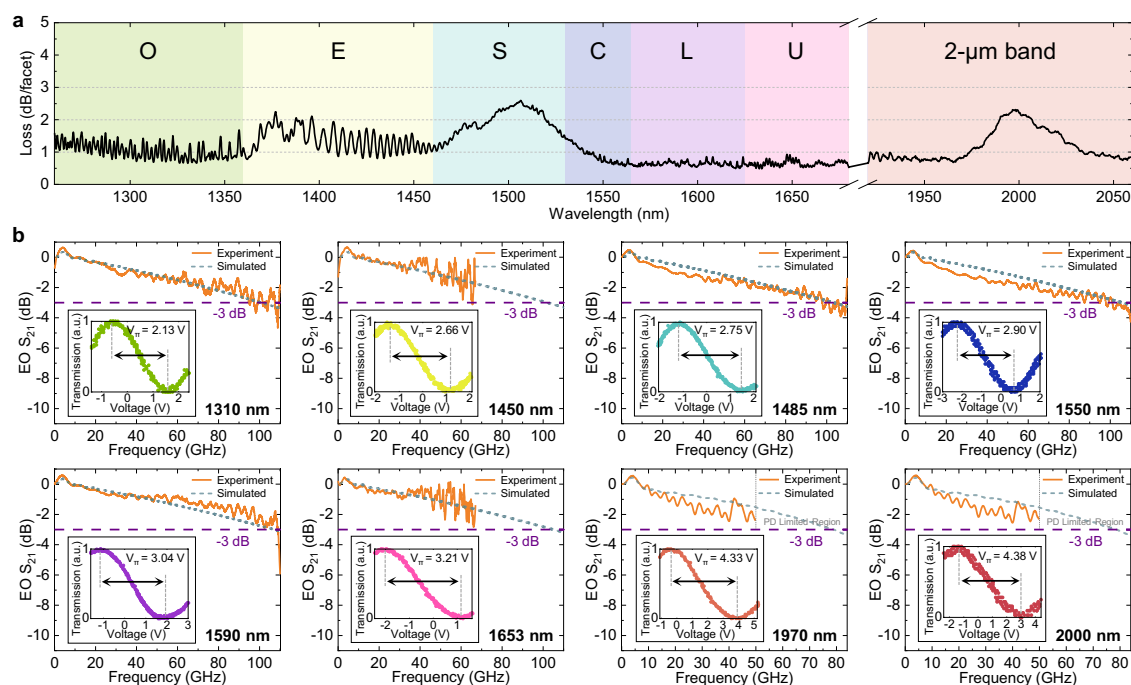


Fig. 3 | Measured performance of the thin-film lithium niobate (TFLN) modulator. **a** Measured coupling loss of the spot-size converter (SSC) for the 1260–1680 nm and 1920–2060 nm range. **b** Measured electro-optic (EO) S_{21}

responses of the TFLN modulator. The dash lines indicate the simulated bandwidths. The insets show normalized optical transmissions with measured half-wave voltage V_{π} .

of on-chip loss was identified as absorption by the metal electrodes, which becomes more pronounced at longer wavelengths (e.g., 1970 nm). Detailed loss analysis of the fabricated modulator can be found in Supplementary Note 6. Experimental on-chip losses for a gap spacing ($G = 6 \mu\text{m}$) show strong agreement with simulated values, as illustrated by the dashed lines in Fig. 2f.

Figure 3b illustrates the half-wave voltage V_{π} measurements with a 100 kHz triangular voltage sweep. With a modulator length of 9 nm, the $V_{\pi} \cdot L$ values are calculated to be 1.92/2.39/2.48/2.61/2.74/2.89/3.90/3.94 V·cm for $\lambda = 1310/1450/1485/1550/1590/1653/1970/2000$ nm, respectively. The simulated and measured $V_{\pi} \cdot L$ parameters across these eight wavelengths can be found in Supplementary Note 7. Notably, the experimental $V_{\pi} \cdot L$ and loss are consistent with the simulation results in Fig. 2f for $G = 6 \mu\text{m}$, validating the accuracy of our numerical model. This agreement confirms the reliability of our simulations for predicting performance trends across varying values of electrode gap. For instance, the simulations suggest that prioritizing reduced metal absorption loss over minimizing V_{π} can be achieved by increasing G slightly to $7 \mu\text{m}$ would maintain comparable high-frequency EO performance. Additionally, static extinction ratios (ERs) were measured by sweeping the heater voltage, yielding values of -17, -15, and -18 dB at 1310, 1550, and 1970 nm, respectively. The observed wavelength-dependent ER variations primarily originate from power imbalance between the output ports of the 3-dB splitter, attributed to fabrication tolerances.

The EO responses of the fabricated TFLN modulator were characterized using an 110-GHz lightwave component analyzer (LCA, Ceyear 6433 P) in conjunction with a tunable laser (Santec TSL-570). A 110-GHz high-speed RF probe was employed to deliver RF signals from the LCA, while a 38- Ω load resistor was attached at the end of the traveling-wave electrode. As shown in Fig. 3b, the 3-dB bandwidths of ~100 GHz were observed at wavelengths of 1310 nm (O-band), 1485 nm (S-band), 1550 nm (C-band), and 1590 nm (L-band). Due to the lack of relevant wavebands detection modules in the LCA, we utilized a 67-GHz vector network analyzer (VNA, Keysight N5247B) to measure the

EO S_{21} response. The EO response was captured by feeding electrical signals from the PDs into the VNA. The measured EO response curves for the E- (1450 nm) and U-bands (1653 nm) using the commercial PD revealed that the modulator's 3-dB bandwidth exceeded 67 GHz, limited by the VNA's bandwidth. For the 2- μm band, where commercial PDs lack sufficient bandwidth, we employed a recently demonstrated high-speed GeSn PD (bandwidth >40 GHz, ref. 16) to measure the bandwidth. The details of GeSn PD and the EO frequency responses for the 2- μm band measurement are shown in Supplementary Note 8. After system calibration and PD-response de-embedding, characterization of the frequency response and high-speed data transmission capabilities was achieved. The measured EO S_{21} response at $\lambda = 2 \mu\text{m}$ reveals a 3-dB bandwidth of >50 GHz, the highest reported to date for optical modulators operating in the 2- μm band. As further indicated by simulations (Fig. 3b, dashed lines), this bandwidth could be extended to ~80 GHz. The low-frequency peaks observed in both the simulated and measured responses stem from the deliberate impedance configuration ($Z_L < Z_0$), which was implemented to increase bandwidth by compensating the drop-off at low frequencies³⁷.

To evaluate the high-speed data transmission performance of the TFLN modulator, the device was optically packaged to UHNA4 fibers and characterized across all operational wavelength regions: the O-band (1310 nm), E-band (1450 nm), S-band (1485 nm), C-band (1550 nm), L-band (1590 nm), U-band (1653 nm), and 2- μm band (1970/2000 nm). The selected wavelengths in each band were chosen to align with current and potential communication applications, as well as the available laser sources and amplifiers. Figure 4a illustrates the experimental setup for measuring eye diagrams and BER. Optical signals were generated using various laser sources at 1310 nm, 1450 nm, 1485 nm, 1550 nm, 1590 nm, 1653 nm, 1970 nm, and 2000 nm. High-frequency RF signals, synthesized by a 256 GSa/s arbitrary waveform generator (AWG, Keysight M8199B), were amplified to $\sim 2.4 V_{pp}$ using an RF amplifier and applied to the modulator. A pre-equalizer was implemented to preprocess signals prior to loading them into the AWG, thereby compensating for bandwidth limitations

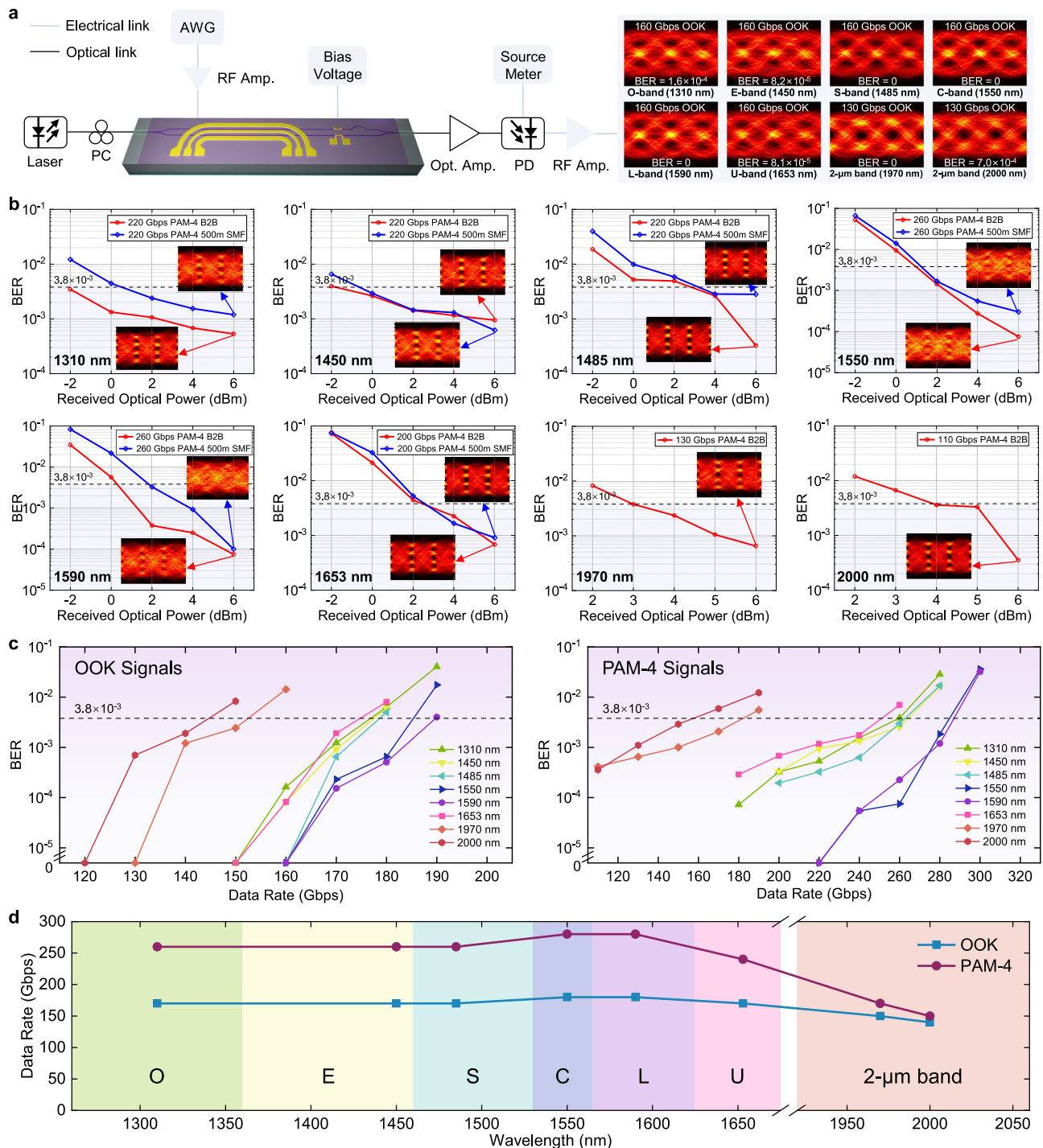


Fig. 4 | Data transmission results. **a** Experimental setup for measuring the eye diagrams and the calculated eye diagrams for On-Off Keying (OOK) signals.

b Measured bit error rate (BER) curves versus the received optical power of four-level pulse amplitude modulation (PAM-4) signals for $\lambda = 1310/1450/1485/1550/1590/1653/1970/2000$ nm. The insets show the calculated eye diagrams. **c** The BER

curves with increasing data rates of different wavelengths with OOK and PAM-4 signals. **d** The maximum data rates of OOK/PAM-4 signals with BERs under the hard-decision forward error correction (HD-FEC) threshold operated in the ultra-wide wavebands. AWG arbitrary waveform generator, PC polarization controller, RF Amp. RF amplifier, Opt. Amp. optical amplifier, PD photodetector.

of the system. The bias point of the modulator was stabilized via an integrated heater. The modulated light was amplified using wavelength-specific amplifiers: semiconductor optical amplifiers (SOAs) for the O-, E-, S-, and U-bands; two erbium-doped fiber amplifiers (EDFAs) for the C- and L-bands; and a thulium-doped fiber amplifier (TDFA) for the 2- μ m band. The amplified light was then coupled into a 100-GHz-bandwidth PD (Finisar XPDV4121R) for

wavelengths up to the U-band, and a GeSn PD (previously characterized in prior work) for the 2- μ m band. Received signals were captured by a high-speed real-time oscilloscope (Keysight UXRO594BP), featuring a 59-GHz bandwidth and 256 Gs/s sampling rate. The BER can be calculated after a series of offline digital signal processing (DSP). A Volterra nonlinear equalizer (VNLE) and maximum likelihood sequence equalization (MLSE) algorithm was jointly employed to

mitigate both linear and nonlinear signal impairments. Detailed specifications of the equipment used in high-speed data transmission measurements are provided in Supplementary Note 9.

The inset of Fig. 4a displays measured eye diagrams and corresponding BERs for 160 Gbps OOK signals across the O-, E-, S-, C-, L-, and U-bands, as well as 130 Gbps OOK signals in the 2- μ m band. Clear eye openings were observed for all the wavelengths, respectively. To our knowledge, this represents the highest OOK transmission rate in the 2- μ m band and the first TFLN modulator capable of exceeding 170 Gbps OOK operation across all O-U bands. We further evaluated PAM-4 performance by measuring BER curves as a function of received optical power (ROP), defined as the optical power incident on the PD. A variable optical attenuator (VOA) was inserted after the optical amplifiers to adjust ROP. Figure 4b shows back-to-back (B2B) BER curves for the TFLN modulator. For O-U bands, transmission results through a 500-m standard SMF were also included and exhibit negligible power penalties compared to the B2B case. To further explain the data rates of single-lane transmission, we analyze the BER performance of OOK and PAM-4 signals for all eight wavelengths ranging from 120 to 300 Gbps (see Fig. 4c). Under the same HD-FEC threshold (3.8×10^{-3}), we successfully realize the transmission of 260/260/260/280/280/240/170/150 Gbps PAM-4 signals were obtained at 1310/1450/1484/1550/1590/1653/1970/2000 nm. Figure 4d shows the maximum data rates achieved using OOK/PAM-4 signals while maintaining a BER below the HD-FEC threshold for all eight wavelengths. This represents the first experimental validation of full-spectrum optical communications (O to U bands) with single-lane data rates exceeding 240 Gbps (PAM-4), enabled by the modulator's ultra-wide wavelength operability. Furthermore, the achieved single-lane 170 Gbps transmission in the 2- μ m band significantly surpasses the previously reported record for this spectral region²². Further discussions on data transmission measurements for various wavebands can be found in Supplementary Note 10.

Discussion

Figure 5 compares the 3-dB EO bandwidths of integrated modulators in the short-wave infrared spectrum regime from 1.2 to 2.1 μ m^{43–47}.

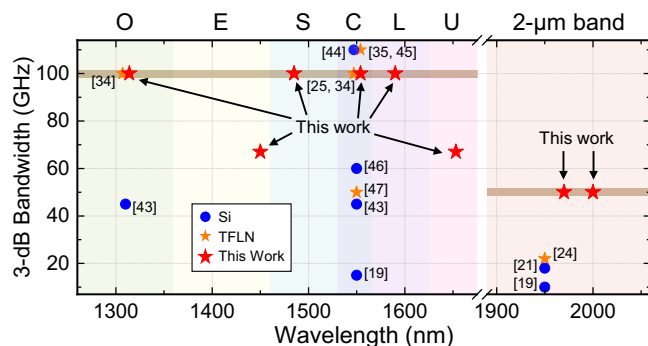


Fig. 5 | Comparisons of integrated modulators. Comparison of 3-dB bandwidth of short-wave infrared high-speed electro-optic (EO) modulators.

Our TFLN modulator operates across an unprecedented 800-nm optical bandwidth, spanning the O-U bands (1260–1680 nm), and extending into the 2- μ m regime (1920–2060 nm). This broad operational range enables multi-band compatibility within a single device, a critical advantage for versatile photonic systems. The TFLN modulator maintains a flat frequency response across its operational range. The measured EO bandwidths of the modulator reach ~100 GHz at critical communication wavelengths of 1310 nm (O-band), 1485 nm (S-band), 1550 nm (C-band), and 1590 nm (L-band), which are comparable with state-of-the-art performance of single-waveband-optimized integrated modulators reported in literature^{25,34,35,44,45}. The 50-GHz bandwidth at 2000 nm enables, to our knowledge, demonstration of the highest single-lane 170 Gbps transmission data transmission in this band. This represents a 2.3-fold improvement in bandwidth over existing 2- μ m modulators, directly enhancing data capacity and underscoring the potential for next-generation high-speed communication systems. The detailed comparisons of the state-of-the-art integrated EO modulators operating in the 2- μ m spectral band are given in “Methods”.

In conclusion, we demonstrate a TFLN modulator featuring high EO bandwidth and a record-breaking operational range spanning over 800 nm, from the O-band to the 2- μ m spectral region. The device achieves 3-dB EO bandwidths of over 67 GHz in the O- to U-bands and 50 GHz in the 2- μ m band. Specifically, the modulator achieves a 3-dB EO bandwidth of ~100 GHz at 1310 nm (O-band), 1485 nm (S-band), 1550 nm (C-band), and 1590 nm (L-band). High-speed data transmissions of over 170 Gbps OOK signals and 240 Gbps PAM-4 signals were experimentally validated at all O-U bands. Furthermore, 150 Gbps OOK signals and 170 Gbps PAM-4 signals were also demonstrated for the 2- μ m band. The BERs are all below the HD-FEC threshold of 3.8×10^{-3} . These results underscore the transformative potential of TFLN modulators in enabling ultrabroadband optical communication systems that seamlessly bridge conventional telecom bands with the emerging 2- μ m window. This advancement directly addresses the escalating bandwidth demands of next-generation data centers and high-performance computing infrastructures.

Methods

Fabrication of the TFLN modulator

The fabrication process of the TFLN modulator is outlined as follows: First, electron beam lithography (EBL) was used to define ridge waveguide structures on a 700-nm-thick layer of AR-P 6200 resist. The ridge waveguide was then formed by etching 180 nm of lithium niobate (LN) using Ar⁺-based inductively coupled plasma (ICP) dry etching. A 2- μ m-thick silica cladding layer was deposited over the waveguide via plasma-enhanced chemical vapor deposition (PECVD) to encapsulate the modulation section. This silica layer was selectively etched using ICP dry etching in the electrode and SSC regions. To create the LN bi-layer tapers, the EBL and ICP etching steps were repeated to remove the remaining 120 nm of LN. A 200-nm-thick thermal titanium (Ti) layer was deposited by electron-beam

Table 2 | Performance comparison of on-chip integrated EO modulators operating at 2- μ m band

Ref.	Platform	Structure	Extinction Ratio (dB)	3-dB Bandwidth (GHz)	Max. Baud Rate and Signal Format	Other wavebands capable
19	SOI	MZI	-	10	20 Gbaud OOK	C-band
20	SOI	Michelson	15	-	20 Gbaud OOK 15 Gbaud PAM-4	No
24	LNOI	MZI	20	22	32 Gbaud OOK	No
21	SOI	MZI	22	18	30 Gbaud OOK 40 Gbaud PAM-4	No
22	SOI	MRM	19	18	50 Gbaud OOK	No
23	SOI	Racetrack Ring	21	26	34 Gbaud OOK	No
This work	LNOI	MZI	18	>50	150 Gbaud OOK 85 Gbaud PAM-4	O-U bands

evaporation (EBE), after which the heater pattern was defined via EBL. A 1- μm -thick Au traveling-wave electrode was then formed by EBE deposition and lift-off. Next, a 4.2- μm -thick SiON layer was grown across the entire chip using PECVD with silane, ammonia, and nitrous oxide precursors. The SiON layer was patterned and etched twice to form the SiON ridge waveguide in the SSC region, while the SiON above the modulation section was removed. A 2- μm -thick protective SiO₂ layer was deposited by PECVD to shield the SSC and minimize contamination-induced losses in the modulator. Finally, the cladding thickness in the modulation section was reduced to 2 μm via ICP etching, and the electrode pads were exposed.

Comparisons of the 2- μm integrated EO modulators

Table 2 summarizes the performance metrics of integrated electro-optic modulators operating in the 2- μm spectral band. Our TFLN modulator achieves a >50 GHz EO 3-dB bandwidth—the highest reported value in this spectral region—enabling a baud rate of 85 Gbaud (170 Gbps for PAM-4) and surpassing prior state-of-the-art devices. This advancement establishes a new benchmark for high-speed data transmission in the 2- μm band.

Data availability

The raw data generated in this study have been deposited in the Figshare database under accession code <https://doi.org/10.6084/m9.figshare.30618413>.

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Author contributions

Q.L., Q.Y., A.P., and L.S. conceived the project. Q.L., Q.Y., and A.P. carried out simulations and designed the TFLN modulator. A.P. and C.S. fabricated the modulator. Q.L. and A.S. conducted the measurements. Y.D., S.X., C.S., and S.Z. assisted in the measurements across the O-U

bands. J.C., Y.Z., J.L., and J. Zheng provided the 2-μm photodetector and assisted in the measurements across 2-μm band. Q.Y., A.S., S.Z., and L.S. analyzed and discussed the experimental results. Q.L. and L.S. wrote the manuscript with contributions from all authors. The project was carried out under the supervision of J. Zheng, J. Zhang, N.C., C.Z., J.X., L.S., and M.Z.

Competing interests

The authors declare no competing interests.

Additional information

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